

Year 8
Mini Test 1
Indices & Integers

Name: _____ Class: 8-__ Teacher: _____

Time Allocation: 25 minutes

Marks: ____/50

(NON- CALCULATOR)

1. Complete the following table (1,1,1,1,1,1,1)

Index Notation	Expanded Notation	Numeral
3^2	3×3	9
5^3	$5 \times 5 \times 5$	125
$2^6, 8^2, 4^3$	$8 \times 8, 4 \times 4 \times 4 \dots$	64
10^6	$10 \times 10 \times 10 \times 10 \times 10 \times 10$	1000000

2. Use index laws to solve the following (1,1,1,1,1,1,2,2,2)

a) $4^3 \times 4^2 =$

$4^{3+2} = 4^5$

c) $a^2 \times a^7 =$

$a^{2+7} = a^9$

b) $10^5 \div 10^2 =$

10^3

d) $\frac{m^4}{m^2} =$

$m^{4-2} = m^2$

e) $(b^4)^5 =$

b^{20}

f) $(3^6)^9 =$

3^{54}

g) $x^0 =$

1

h) $3x^0 =$

3

i) $(7x)^2 =$

$49x^2$

j) $\frac{21b^8}{6b^3} =$

$\frac{7b^5}{2}$

3. Use index laws to solve the following (2,2,3)

a) $2^9 \times 2^9 \div 2^2 =$

$2^{18-2} = 2^{16}$

b) $3a^6 \times 2a^3 \div a^2 =$

$6a^9 \div a^2 = 6a^7$

c) $(t^4)^2 \times 9t^3 \div 3t^2 =$

$t^8 \times 3t^1 = 3t^9$

4. Show that anything to the power of zero equals one. (3)

repeated division
Index Law 2
Division fact.

$$\begin{aligned} 2^2 &= 4 \downarrow \div 2 \text{ (the base)} \\ 2^1 &= 2 \downarrow \div 2 \\ 2^0 &= 1 \downarrow \div 2 \end{aligned}$$

$$\begin{aligned} 3^2 &= 9 \downarrow \div 3 \\ 3^1 &= 3 \downarrow \div 3 \\ 3^0 &= 1 \downarrow \div 3 \end{aligned}$$

5. Solve the following. (1,1,1,1,2,1,1,1,1,4)

a. $14 - 5 = 9$

b. $6 - - 5 = 11$

c. $-3 + 4 = 1$

d. $-5 - 8 = -13$

e. $3 - 8 + (-2) = -7$

f. $-27 + 13 = -14$

g. $4 \times 5 = 20$

h. $-6 \times -4 = 24$

i. $-15 \div 5 = -3$

j. $4 \times (-2) \times 7 = -56$

k. $[(2^3 \times 2) \div (2^2 \div 2)] \times 3 =$
 $2^4 \div 2 \times 3 = 2^3 \times 3 = 8 \times 3 = 24$

6. Solve the following word problems (1,1,2)

a) The product of two numbers is (-24) and their sum is (-5), What are the two numbers?

$-4 \times 6 = -24$
 $-8 \times 3 = -24$
 $-3 \times 2 = -6$
 $(-8, 3)$

b) The average of three numbers is (-4), A new number is added to the list making the average (-3). What is the new number?

$a + b + c = x/2 = -4$
 $(+3)$

c) The temperature in a mountain hut is 15°C at 9.00pm on Monday night. It drops by 2°C per hour for 11 hours, and then the next morning, rises by 1°C per hour for the next four hours. What is the temperature at midday on Tuesday?

15°
 $- 22$
 $= -7^\circ$
 $+ 4$
 $= -3^\circ$

-3°C

THE END